

Educational Enrollment Analysis Across Selected Countries (2016–2019)

Introduction

In this project, [World Bank Open Data](#) dataset was used to perform analysis on educational enrollment.

The purpose of this study was to examine and compare the student enrollment at various education levels in a number of selected countries; Qatar, Jordan, Oman, China, India, and Georgia, over the period 2016 – 2019.

Dataset Description

The dataset consists of the number of enrollments in education for three key education indicators:

- Enrolment in primary education
- Enrolment in secondary education
- Enrolment in tertiary education

Countries used in the analysis:

Qatar , Jordan , Oman , China , India , Georgia

Time period:

2016-2019

Preprocessing

To guarantee quality and consistency of the data, the following procedures were applied:

- **Numeric Conversion:** Numeric columns were converted to whole numbers.
- **Custom Imputation:** To keep geographical data integrity, missing values in numerical columns were imputed with the mean of each country (row).
- **Data Restructuring:** The multi year columns were concatenated into one column to allow for time series analysis in Power BI.

Dashboard Features

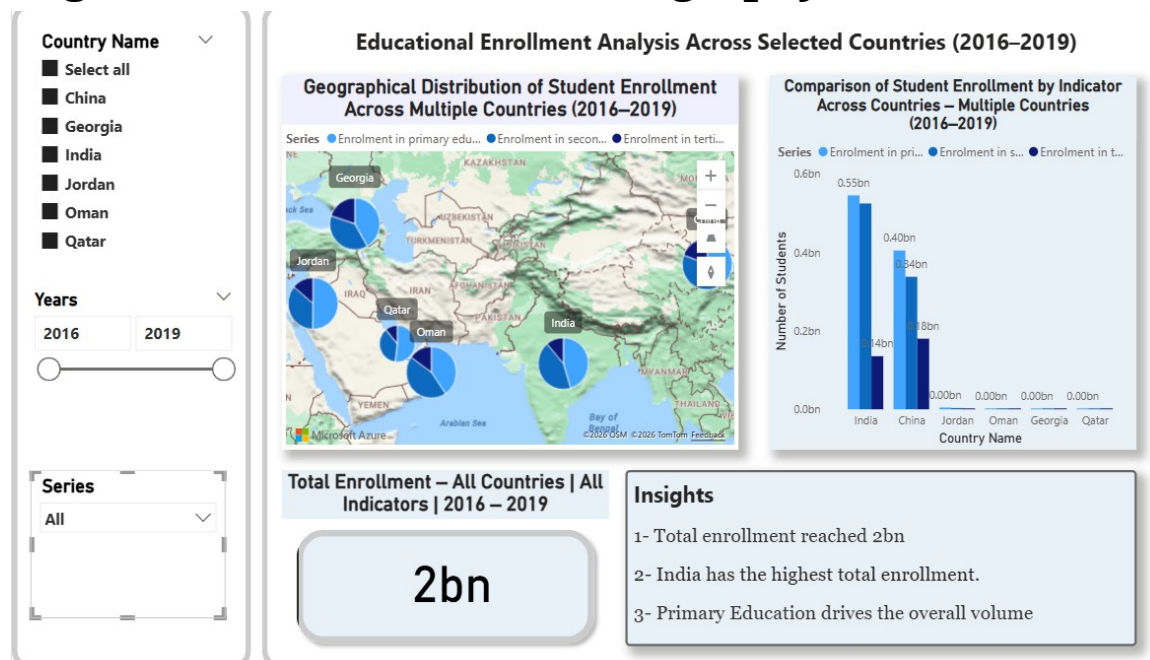
- **Dynamic Chart Titles (DAX Measure):** In the charts themselves, instead of a static text, we created dynamic titles using a DAX measure. These titles are updated when slicer selections are applied (Country, Year, Level) so that the user can know precisely what data is being plotted.
- **Basic Filtering Strategy:**
 1. **Filters on this page (Years):** Used to restrict the analysis to particular time periods within a single page.

2. **Filters on all pages (Series):** Applied to keep selected level of education checked as user moves across dashboard.
- **Interactive Slicers:** The dashboard contains three main slicers (Years, Country, Series) which allow users to customize the view and analyze the data chunks in more detailed

Visualization

We created a two-page interactive dashboard in Power BI to study global education enrolment:

Page 1: Global Overview & Geography

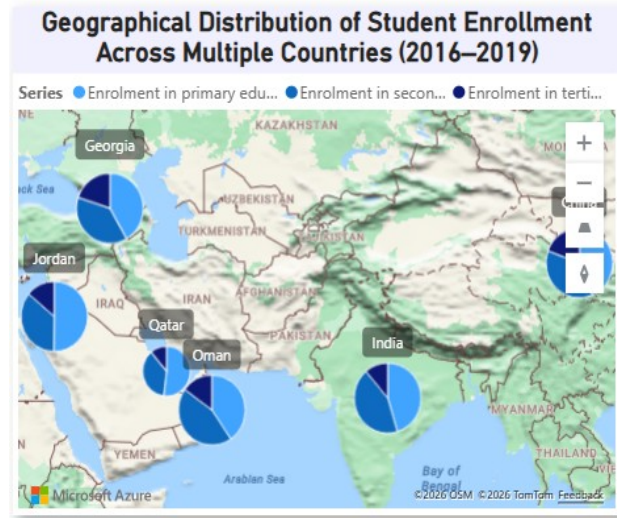


Explanation of the charts used:

1- Bubble Map

It displays the geographical student enrollment of the selected countries.

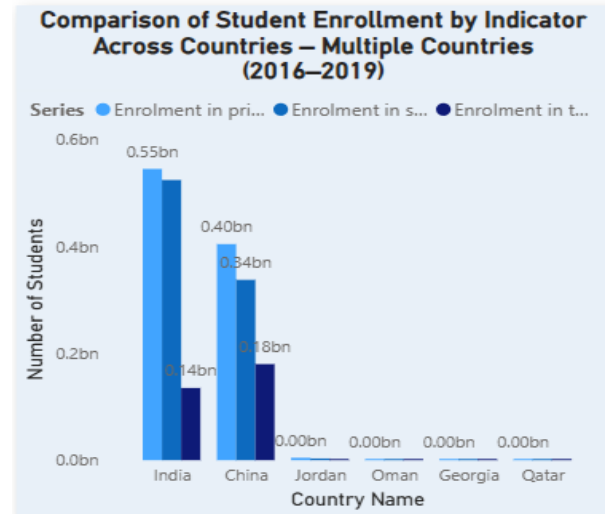
The bubble size indicates the number of students, and the color indicates the Education Series (Primary, Secondary, or Tertiary). When the user selects a particular country, year or education level, the map automatically updates, enabling a neat spatial comparison



2- Column Chart:

It displays the number of students enrolled in various levels of education in a chosen country and year.

Consistent use of color for each Series enables the reader to quickly determine the highest and lowest levels of enrolment at a glance among the different countries.



3- Data Card :

It shows a single value that is a summary, which is the total number of students for the applied filters.

This card gives a high level overview with the users able to know the total number of enrollments at a glance without having to study down to the

Total Enrollment – All Countries | All Indicators | 2016 – 2019

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4- Slicers:

To choose a specific country or a specific year or indicator .

Country Name

Select all

China

Georgia

India

Jordan

Oman

Qatar

Years

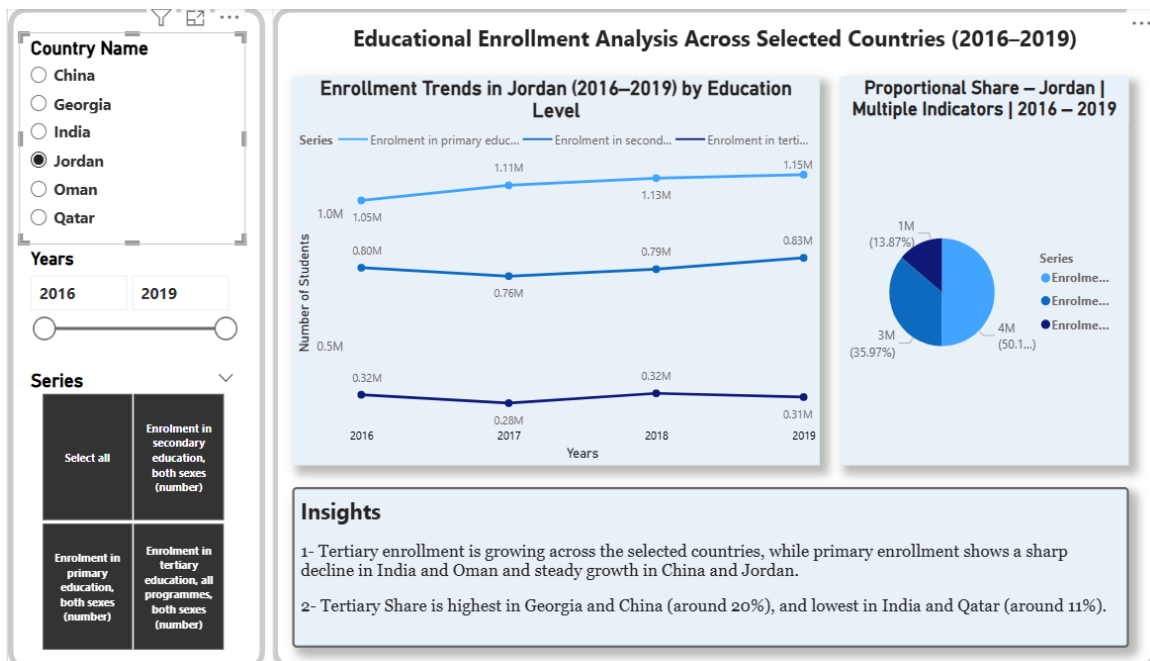
2016

2019

Series

All

Page 2: Trend & Distribution Analysis



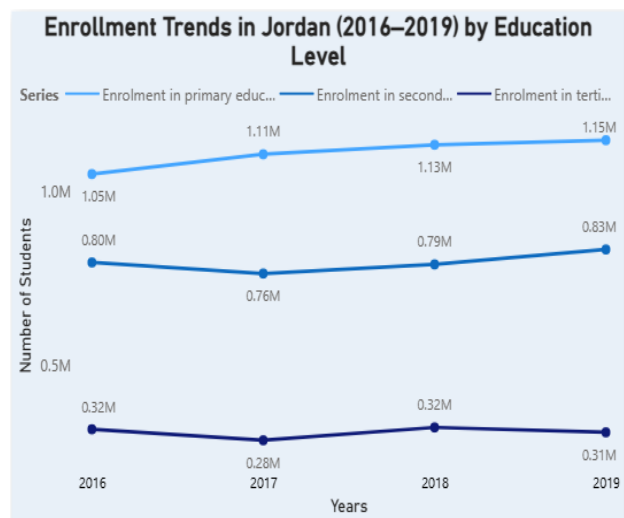
Explanation of the charts used:

1- Line chart :

It displays the changes in the number of students who attended three levels of education, namely primary, secondary and tertiary, from 2016 to 2019.

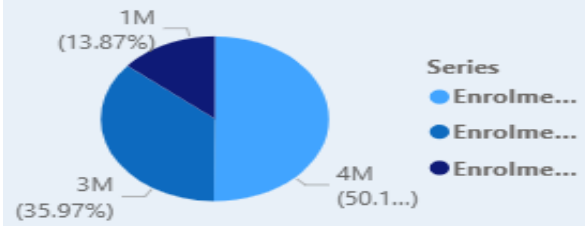
It shows the relative proportion of students in each of the three levels of education. This chart is a quick glance at how the total enrollment in education is distributed among levels (primary, secondary and tertiary).

Displaying the share of each category, it immediately allows the user to determine which is the most popular education level in



2- Pie chart:

Proportional Share – Jordan | Multiple Indicators | 2016 – 2019



3- Slicers:

To choose a specific country or a specific year or indicator .

Country Name

☐ China

☐ Georgia

☐ India

☒ Jordan

☐ Oman

☐ Qatar

Years

2016 2019

Series

Select all

Enrolment in secondary education, both sexes (number)

Enrolment in primary education, both sexes (number)

Enrolment in tertiary education, all programmes, both sexes (number)

Summary:

Based on the data, enrollment patterns vary considerably across the six selected countries. India has the highest total enrollment, while China records the highest tertiary enrollment.

Tertiary enrollment is growing across the selected countries. Primary enrollment shows a sharp decline in India and Oman and steady growth in China and Jordan. Georgia and China have the highest tertiary enrollment shares at approximately 20%, while India and Qatar have the lowest at approximately 11%.

Overall, the dashboard highlights clear differences in enrollment trends and distribution across countries and education levels between 2016 and 2019.

Conclusion:

Power BI enabled us to transform a complex educational dataset into a clear and interactive dashboard. The dashboard effectively presents enrollment trends and supports comparisons across countries, education levels, and years. Through dynamic titles, filters, and visualizations, users can efficiently explore the data and identify key enrollment patterns across the selected countries between 2016 and 2019.